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ABSTRACT

Purpose: To analyse the effect of resistance training (RT) performed until volitional failure with low-, moderate- and high-loads on muscle hypertrophy and muscle strength in healthy adults; and assess the possible participant-, design-, and training-related covariates which may affect the adaptations. **Methods:** Using PRISMA guidelines, *MEDLINE*, *CINAHL*, *EMBASE*, *SPORTDiscus*, and *Web of Science* databases were searched. Including only studies that performed sets to volitional failure, the effects of low- (>15 RM), moderate- (9-15 RM), and high-load (≤8 RM) RT were examined in healthy adults. Network meta-analysis was undertaken to calculate the standardised mean difference (SMD) between RT loads in overall and subgroup analysis involving studies deemed high-quality. Associations between participant-, design-, and training-related covariates with SMD's were assessed by univariate and multivariate network meta-regression analysis. **Results:** Twenty-eight studies involving 747 healthy adults were included. Although no differences in muscle hypertrophy between RT loads were found in overall (P= .113 - .469) or subgroup analysis (P= .871 - .995), greater effects were observed in untrained participants (P= .033), and participants with some training background who undertook more RT sessions (P= .031 - .045). Muscle strength improvement was superior for both high-load and moderate-load compared to low-load RT in overall and subgroup analysis (SMD= 0.60 – 0.63 and 0.34 – 0.35, respectively; P< .001 - .003), with a non-significant but superior effect for high- compared to moderate-load (SMD= 0.26 - 0.28, P= .068). **Conclusion:** While muscle hypertrophy improvements appear to be load independent, increases in muscle strength are superior in high-load RT programs. Untrained participants exhibit greater muscle hypertrophy while undertaking more RT sessions provides superior gains in those with previous training experience.

Keywords: Strength training; Volitional failure; Muscle hypertrophy; Muscle strength.

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INTRODUCTION

Resistance training is a popular and effective modality to improve muscle function, functional performance, and health parameters in a wide range of healthy and clinical populations. Among the many expected outcomes, increases in muscle size and strength are considered important and desirable by individuals and clinicians either for performance or health and functional improvement. In the 1940s, Thomas DeLorme proposed undertaking resistance exercise sets until neuromuscular volitional failure to maximize such benefits (1). Although a vast body of research work in this area has been published (2-7), issues regarding how to optimize resistance training outcomes remain (8, 9). Furthermore, controversies regarding how volitional failure is operationalised calls into question the implementation of this technique in populations other than strength athletes as participants' motivation and tolerance, discomfort, and neuromuscular fatigue affects the performance and results related to this training program.

Load selection has been considered an important resistance training variable to successfully increase muscle size and strength across different populations (10).Considering Henneman's size principle (i.e., motor units are recruited from smallest to largest) (11), studies have advocated in favour of either high loads (12-15) or both low and high loads (16) to achieve maximal or near-maximal recruitment of motor units during fatiguing contractions to induce muscle hypertrophy. Although this is a topic of intense debate in the literature, when low-load sets are performed until volitional failure, neuromuscular fatigue necessitates increasing percentage recruitment of the motor unit pool, and, through this mechanism (12), such training may produce meaningful drive for muscle hypertrophy. For example, Mitchell et al. (6) and Lim et al. (5) have reported that 10 weeks of resistance training until volitional failure in untrained

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men at low- and high-loads (30% and 80% of 1-RM) resulted in similar increases in quadriceps femoris muscle volume (6.8% and 7.2%, respectively) and muscle fibre cross-sectional area (CSA) of the vastus lateralis (ranging from 15 to 20% in both groups). These findings indicate that muscle hypertrophy may be more responsive in untrained individuals because of the large window for adaptation, masking differential effects of training modalities and dosages (17), and not show an obvious load-dependent relationship when resistance training sets are performed until volitional failure (6, 18). In contrast, Schoenfeld and colleagues (7) reported that 8 weeks of resistance training at high-loads (2-4 RM's) induced greater strength gains in recreationally trained men compared to moderate-loads (8-12 RM's), while increases in elbow extensor and quadriceps femoris muscle thickness were higher for the moderate-load group. Consequently, it is unclear as to loading effects on muscle hypertrophy when resistance training is undertaken until volitional failure. Furthermore, despite previous meta-analyses examining low (≤60% of 1-RM) and higher resistance training load (>60% of 1-RM) effects on muscle strength and hypertrophy (9, 19), the lack of meta-analyses comprising a large number of studies comparing well-defined ranges of load such as low- (<60% of 1-RM), moderate- (between 60 to 79% of 1-RM) and high-load resistance training (≥80% of 1-RM) through robust meta-analytic approaches such as network meta-analysis, precludes determination of an appropriate load for outcomes of interest in healthy adults with different pre-training genetic and morphological characteristics.

Other issues comparing resistance training loads are related to the heterogeneity of study designs such as the participants involved (men vs. women or combined), training status (untrained vs. recreationally trained vs. strength athletes), experimental design (between- and within-subjects), assessed outcomes (lower-, upper- and whole-body) and training prescription

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(number of sessions; operational definition of volitional failure and its implementation and verification). These different characteristics among studies may preclude accurate evaluation of an optimal resistance training load, considering specific methodological or resistance training prescription characteristics when full and similar recruitment of the motor unit pool is achieved. As a result, the purpose of the review and analysis is to: 1) analyse the effect of resistance training performed until failure with low-, moderate- and high-loads on muscle hypertrophy and muscle strength in healthy adults; and 2) assess the possible participant-, design- and training-related covariates which may affect the hypertrophy and strength gains.

METHODS

Study selection procedure

The study was undertaken in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (20, 21), and the method used was based on the minimum criteria established by the Cochrane Back Review Group (CBRG) (22). The review included published data from experimental studies that evaluated the effects of low- (<60% of 1-RM, or >15 RM), moderate- (between 60 to 79% of 1-RM, or 9-15 RM), and high-load (≥80% of 1-RM, or ≤ 8RM) in resistance training performed until volitional failure in healthy adults (23). The primary outcomes of this review were muscle hypertrophy (i.e., defined as a measure of muscle mass or size) and muscle strength (i.e., defined by 1-RM tests). Studies were excluded when: 1) they did not present sufficient information regarding the comparisons between different loads or pre- and post-training values in the resistance training programs; 2) interventions were shorter than 6 weeks; 3) specific outcomes for this review or sufficient information were not reported (e.g., baseline and post-intervention assessment, and within- and

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between-groups mean difference); 4) resistance training involving blood flow restriction protocols; and 5) written in a language other than English. In the search strategy, titles and abstracts were first independently evaluated. When abstracts did not provide sufficient information, they were selected for full-text evaluation. Eligibility was assessed independently by two authors (P.L. and J.T.), with differences resolved by consensus.

The search was conducted up to December 2019 using the following electronic databases: MEDLINE, CINAHL, EMBASE, SPORTDiscus, and Web of Science. The terms used were: 'resistance training until failure' and 'muscle hypertrophy' or 'muscle strength' in association with a list of sensitive terms to search for experimental studies. In addition, a manual search was performed in the reference lists provided in the selected papers as well as in a previous systematic review (9) to detect studies potentially eligible for inclusion. The search strategy used is shown in the Supplemental Digital Content (SDC) Table S1 (see in SDC1, literature search strategy).

Data extraction

The data extraction was performed via a standardised form. Information regarding participants, resistance training protocols, outcomes, and assessment techniques were collected. Study characteristics, intervention duration, number of sessions, sex, experimental design, training status, assessed outcomes, and resistance training prescription method were extracted, along with the main outcomes, while outcomes were extracted in their absolute units (e.g., kg for 1-RM assessments; mm or cm for muscle thickness). When graphs were used instead of

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numerical data, the graphs were measured using a specific tool for data extraction (WebPlotDigitizer, San Francisco, California, USA).

Assessment of Risk of Bias

The risk of bias of individual studies was evaluated according to the 2nd version of the Cochrane risk-of-bias tool for randomized trials (RoB 2) (24), focusing on different aspects of trial design, conduct, and reporting. Each assessment using the RoB 2 tool is focused on the outcome level. The six-item instrument used to evaluate each included RCT in each outcome of interest is as follows: 1) randomisation process; 2) deviation from intended interventions; 3) missing outcome data; 4) measurement of the outcome; 5) selection of the reported result; and 6) overall analysis. Overall risk of bias was expressed as "low risk of bias" if all domains where classified as low risk, "some concerns" if some concern was raised in at least one domain but not classified as at high risk in any other, or "high risk of bias" if at least one domain was classified as high risk, or have multiple domains with some concerns (24).

Data analysis

In the network meta-analysis, the pooled-effect estimates were obtained from the standardised mean difference (SMD) of baseline to the final assessment of the intervention for each group. When studies did not provide standard deviation (SD) of change in the outcomes, these values were estimated using a correlation coefficient (r) of 0.5 and the equation:

$$\mathrm{SD}_{\mathrm{change}} = \sqrt{\mathrm{SD}_{\mathrm{baseline}}^2 + \mathrm{SD}_{\mathrm{final}}^2 - (2r \times \mathrm{SD}_{\mathrm{baseline}} \times \mathrm{SD}_{\mathrm{final}})}$$

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as per the Cochrane Handbook guideline (25). Furthermore, to avoid overestimating the weight of a study by entering it multiple times in the analysis (25), experimental groups from the studies were combined when considered within the same resistance training load group (e.g., 3 sets of 30-40 RM and 3 sets of 100-150 RM defined as low load resistance training (2)), as well as outcomes when considered within the same outcome category (e.g., lower-body muscle hypertrophy, or upper-body muscle strength). Analyses were conducted for overall studies, and a subgroup analysis was provided for best quality studies based on the risk of bias assessment. The network meta-analysis was performed following the current PRISMA guideline items (26, 27): 1) a network geometry was created to explore the comparisons between resistance training loads; 2) transitivity was tested by fitting a network inconsistency assumption along with Q test and side-splitting analyses between indirect and direct comparisons; and 3) effect size (ES) was generated considering the heterogeneity and the inconsistency level in the models. Statistical significance was assumed when the SMD reached an α value $\leq .05$. According to Cohen (28), ES values of 0.0 to <0.5 indicate small, 0.51 to <0.8 indicate medium, and ≥0.8 indicate large effects. Furthermore, an estimation of the probability to be the best resistance training program for the outcome was provided based on the consistency values and expressed as a percentage. Outliers were explored by sensitivity analyses omitting one study at a time, generating pooled estimates, and comparing with the original estimates. To check for the presence of publication bias, a network funnel plot and the Egger's test were used, with a significant publication bias considered if the P-value was <0.1 (29). The network meta-analysis was conducted using R (R Core Team, 2019) with the package *netmeta* (30). Forest plots presented for the outcome measures are after sensitivity analysis adjustments.

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To test the association between SMD's and specific covariates such as year of publication, experimental design (i.e., between-subject vs. within-subject), sex (i.e., women vs. men), training status (i.e., untrained vs. recreationally trained), number of sessions, assessed outcomes (i.e., lower- vs. upper-body outcomes) and the prescription method (i.e., %1-RM vs. RM's), univariate and multivariate network meta-regression was employed (31). Dichotomy variables were coded as 0 and 1, while continuous variables were used in the model to explain the variations in muscle hypertrophy and strength among all comparisons. The network meta-regression was conducted using Stata 14.0 with the package *mvmeta* (31).

RESULTS

Studies included

All studies selected reported the aim to compare the effect of different resistance training loads (i.e. low, moderate, or high) on muscle hypertrophy and strength in healthy men and women. We retrieved 5,924 studies, 2,629 of which were retained for screening after duplicate removals. Of these, 2,515 were excluded, and 114 full-text articles were assessed for eligibility (Figure 1). The eligibility assessment resulted in a total of 28 (2-7, 32-53) studies included in the present review, network meta-analyses and meta-regression of which 24 studies (3-7, 32-40, 42-53) examined muscle hypertrophy and 23 studies (2-7, 32, 33, 35, 38-48, 50-52) examined muscle strength. During the eligibility assessment, one of the authors from the studies of Au et al. (54) and Morton et al. (43) was contacted for further information and it was confirmed that the study of Au et al. (54) was a follow-up analysis from Morton et al. (43). As a result, only the study of Morton et al. (43) was included in the systematic review.

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Participants and intervention characteristics

A total of 747 healthy men and women with an average age of 23.4 ± 3.0 years participated in the included studies. Seventeen studies compared low- vs. high-load resistance training (2, 5, 6, 33, 34, 36, 38, 39, 42-46, 49-51, 53), 4 compared low- vs. moderate-load (35, 40, 47, 52), 5 compared moderate- vs. high-load (7, 32, 37, 41, 48), and 2 studies compared low- vs. moderate- vs. high-load (3, 4). Most of the studies involved men (19 of 28 (2-7, 34, 38-42, 44-48, 53), 67.9%) and untrained participants (21 of 28 (2-6, 32-39, 42, 44-46, 49-52), 75.0%) (see in Table S2, SDC2, characteristics of included studies). None of the studies included highly strength trained individuals as defined by 1-RM test values reported in studies involving elite athletes (average relative strength of 2.0 body weights (BW) for back squat and 1.5 BW for bench press (55, 56).

The mean exercise intervention duration was 8.9 ± 2.1 weeks, with an average of 24.6 ± 7.5 sessions (range: 16 to 48). Most of the studies undertook a between-subject experimental design (22 of 28 (2, 3, 5, 7, 32-38, 40-43, 46-49, 51-53), 78.6%), and prescribed the resistance training program by repetitions maximum (18 of 28 (2, 3, 7, 32, 35-37, 40-43, 46-49, 51-53), 64.3%) (see in Table S2, SDC2, characteristics of included studies). Regarding the assessments, muscle hypertrophy was assessed for the lower-limbs in 15 studies (3-7, 32, 35-39, 43, 44, 49, 50), followed by 8 studies assessing upper-limbs (4, 7, 32, 34, 45, 47, 48, 50), and 5 studies assessing whole-body (e.g., dual-energy X-ray absorptiometry (DEXA)) (33, 40, 42, 46, 53), while lower-body muscle strength was assessed in 20 studies (3-7, 32, 33, 35, 38-44, 46, 48, 50-52), followed by 12 studies assessing upper-body muscle strength (2, 4, 7, 33, 40, 41, 43, 45, 47, 48, 50, 51) all using the 1-RM test. Eighteen studies reported the total volume performed during

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the intervention (4, 5, 7, 32-35, 37-45, 48, 50). The number of studies among the resistance training loads for muscle hypertrophy and muscle strength is shown in the network geometry (Figure 2, panel A and B, respectively).

Risk of bias assessment

For muscle hypertrophy, 87.5% of the studies have *some concern* (21 of 24 studies (3-7, 33-39, 43-50, 53)), while the remaining have a *high risk* (3 of 24 studies (32, 40, 42), 12.5%) in the overall risk of bias assessment (Table 1). The concerns in muscle hypertrophy assessment were mainly due to the *randomisation process* as studies did not report concealment allocation (*some concerns*: 23 of 24 studies (3-7, 32-40, 43-50, 53), 95.8%) or did not follow any randomisation (*high risk:* 1 of 24 studies (42), 4.2%). Furthermore, studies were considered to have *some concerns* in the *measurement of the outcome* (8 of 24 studies (4, 7, 38, 39, 44, 47, 48, 50), 33.3%) when evaluating muscle hypertrophy through unblinded evaluations and evaluations requiring technician or assessor direct analysis and interpretation (e.g., muscle-ultrasound imaging), while they had a *high risk* (3 of 24 studies (32, 40, 42), 12.5%) when they did not use a reliable technique of assessment (e.g., skinfold or circumference). In the subgroup analysis for muscle hypertrophy, best quality studies were those considered with *low risk* on the *measurement of the outcome* (3, 5, 6, 33-37, 43, 45, 46, 49, 53).

In the muscle strength overall risk of bias assessment, 87.0% of the studies have *some concern* (20 of 23 studies (3-7, 32, 33, 35, 38-41, 43-48, 50, 52)), while the remaining have a *high risk* (3 of 23 studies (2, 42, 51), 13.0%) (Table 1). The concerns were mainly due to the *randomisation process* as studies did not report concealment allocation (*some concerns*: 20 of 23

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studies (3-7, 32, 33, 35, 38-41, 43-48, 50, 52), 87.0%) or if the participants were not randomly assigned in the experimental groups (*high risk*: 3 of 23 studies (2, 42, 51), 13.0%), and on the *measurement of the outcome* (*some concerns*: 23 of 23 studies (2-7, 32, 33, 35, 38-48, 50-52), 100%) as studies assessed muscle strength with no blinding of testers. In the subgroup analysis for muscle strength, best quality studies were considered those not presenting *high risk* in overall risk of bias assessment (3-7, 32, 33, 35, 38-41, 43-48, 50, 52). The individual risk of bias assessment is shown in SDC Figure S1 panel A and B (SDC3, individual risk of bias assessment).

Resistance training load effects on muscle hypertrophy

Thirty-five comparisons were undertaken on muscle hypertrophy involving 24 studies (3-7, 32-40, 42-53). The results from the consistency network meta-analysis provided no differences in muscle hypertrophy between high- vs. low-load, moderate- vs. low-load or high- vs. moderate-load resistance training (P= .113 - .469; Table 2 and Figure 3). The heterogeneity was I^2= 0%. Furthermore, no differences between the loads were observed in the subgroup analysis involving the best quality studies (number of comparisons= 16; I^2= 0%; P= .871 - .995, Table 2). Although the results of the consistency model indicate that moderate-load (84.5%) and high-load resistance training (75.8%) are the best load for muscle hypertrophy in overall and high-quality subgroup analysis, respectively, the ES's were unlikely to be considered meaningful (small ES, range: -0.09 to 0.15). The inconsistency between direct and indirect comparisons was not significant in the network analysis for all studies (Q= 6.2, P= .103), or in the subgroup analysis (Q= 0.3, P= .957), as well as in the node-splitting analysis across comparisons between

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load groups (all studies: P= .424 - .914; best quality studies: P= .615 - .760). No publication bias was identified after the inspection of funnel plots asymmetry by Egger's test (P= .497 - .909).

In the univariate network meta-regression, the covariates (i.e., year of publication, experimental design, sex, training status, number of sessions, assessed limb, and prescription method) did not explain the variation in muscle hypertrophy gains (P= .278 - .986; see in SDC Table S3, SDC4, univariate meta-regression results). Regarding the multivariate model, there was a significant interaction for training status (P= .033) with untrained participants presenting higher effects on muscle hypertrophy in low- vs. high load resistance training comparison (Table 3). The number of sessions appears to influence the results in both moderate- vs. low-load (P= .031) and high- vs. moderate-load (P= .045) resistance training with a higher number of sessions resulting in greater effects on muscle hypertrophy (Table 3). The remaining variables (i.e., year of publication, experimental design, sex, assessed limb, and prescription method) were not significant in explaining variations in muscle hypertrophy (P= .145 to .999).

Resistance training load effects on muscle strength

Thirty-nine comparisons were undertaken on muscle strength assessed by a 1-RM test involving 23 studies (2-7, 32, 33, 35, 38-48, 50-52). Visual inspection of funnel plots indicated a presence of publication bias (P= .014), and the study of Anderson et al. (2) was considered an outlier in the analysis. After the adjustment, the consistency network meta-analysis results show that high-load (SMD= 0.60, 95% CI: 0.38 to 0.82) and moderate-load (SMD= 0.34, 95% CI: 0.05 to 0.62) resulted in higher muscle strength effects when compared to low-load resistance training (P= <.001 and .003, respectively). A non-significant effect (P= .068) was found

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favouring high-load when compared to moderate-load resistance training (Table 2 and Figure 4). The heterogeneity was I^2= 39.9%, with no presence of publication bias (P= .277). The results of the consistency models indicate that high-load resistance training has a probability of 98.2% to induce greater effects on muscle strength, also sustained in subgroup analyses (k= 36; Table 2) with heterogeneity I^2= 43.7% and no effect of publication bias (P= .350). The inconsistency between direct and indirect comparisons was not significant in the network analysis for all studies (Q= 5.3, P= .150) and for subgroup analysis (Q= 4.4, P= .219), as well as in the node-splitting analyses (all studies: P= .588 - .892; best quality studies: P= .674 - .871).

In the univariate network meta-regression, older studies accounted for the variation in muscle strength in the overall analysis (P= .021), while the remaining variables did not explain variation in muscle strength (i.e., experimental design, sex, training status, number of sessions, assessed limb, and prescription method; P= .097 - .984; see in SDC Table S3, SDC4, univariate meta-regression results). Regarding the multivariate model, older studies (P= .023) and those with men (P= .037) presented higher effects on muscle strength in the low- vs. high-load resistance training comparison (Table 3). The remaining variables did not explain variations in muscle strength in the comparisons (i.e., experimental design, training status, number of sessions, assessed limb, and prescription method; P= .121 to .823, Table 3).

DISCUSSION

There are three important findings from our systematic review and network meta-analysis of the dose-response relationship between resistance training load and gains in muscle hypertrophy and strength. First, in untrained and recreationally trained individuals (i.e., not

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strength athletes), muscle hypertrophy gains are likely to be similar regardless of resistance training load when performed to volitional failure over relatively short periods of intervention. In line with the physiological adaptation principle of *diminishing returns*, untrained participants exhibit greater muscle hypertrophy than those with even modest prior experience in resistance training. Further, undertaking more resistance training sessions provides superior muscle size gains in those with previous training experience. Second, effects on muscle strength are load-dependent, with higher loads resulting in greater gains over the relatively short interventions reviewed. Finally, the results for muscle hypertrophy and strength were maintained even when only the higher quality studies were considered, that is, studies with less risk of bias. Therefore, while improvements in muscle hypertrophy appear to be load independent for untrained and recreationally trained individuals, muscle strength increases are superior with high-load resistance training programs of short duration.

The inclusion of experimental studies employing repetitions until volitional failure is based on ensuring that a similar stressful stimulus was undertaken by all participants in the resistance training programs (57). Although performing resistance training until volitional failure is one way to eliminate a large amount of individual variability related to between-subject endurance capacity (i.e., individuals who are able to complete different numbers of repetitions at a given relative load), and ensures methodological feasibility when comparing different resistance training loads (57), other approaches have been successfully used such as RM zones and percentage of 1-RM, where failure is not mandated but monitored to stay within a training range. Therefore, although exercising until volitional failure is not mandatory for neuromuscular

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adaptations (58, 59), it was the strategy required to compare the efficacy of different resistance training load protocols (57).

Our findings are that performing as many repetitions as possible per set with different loads over relatively short interventions leads to similar muscle hypertrophy in individuals with none or moderate resistance training experience. It appears that any training load can produce a similar magnitude of muscle hypertrophy for different participants (men and women) and muscles assessed (lower- and upper-body). Thus, over relatively short training interventions in untrained or novice subjects, sets to failure is one strategy for gaining muscle hypertrophy, regardless of the load undertaken in resistance training. Other strategies, albeit beyond the scope of this review, if performed with a load that activates a high percentage of motor units', hypertrophy is likely to occur. However, it is important to note that the practical application may still favour the use of a lower number of repetitions using moderate to high loads as the performance of low loads until volitional failure results in higher discomfort due to the higher number of repetitions, longer time under muscular tension and time required (60).

As revealed by the meta-regression, muscle hypertrophy derived from low-, moderate- and high-load resistance training regimes, despite the modest and non-significant difference between them, seem to be affected by training status and number of sessions completed, i.e., volume. Untrained participants exhibit a greater magnitude of muscle hypertrophy compared to those recreationally trained. As reported in previous literature, trained muscles may already present with an increased cross-sectional area (61) and lower anabolic signalling as observed in reduced AMPK and Akt phosphorylation after a resistance training session (62), resulting in an

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attenuated hypertrophic response in participants with previous resistance training experience. However, our findings also indicate greater muscle hypertrophy in recreationally trained participants who undertake a higher number of sessions compared to those undertaking less. It appears that participants with longer experience in resistance training (range: 2 to 7 yrs) (7, 40, 41, 43, 47, 48, 53), while still not highly trained, require a higher volume of training to produce the same or greater hypertrophic adaptation exhibiting the principle of *diminishing returns*. Thus, although there were no differences between training loads for muscle hypertrophy, the meta-regression results suggest that novice participants are likely to experience superior gains than those with previous experience in resistance training undertaking the same volume, while a greater number of sessions may provide additional muscle hypertrophy for those with experience in resistance training. Given the lack of studies with more than 14 weeks duration, any interpretation of these results for longer periods should be viewed with caution.

In contrast to muscle hypertrophy, muscle strength, as defined by 1-RM testing, was found to be dependent on high loads. This finding is expected following: 1) the principle of *specificity* (13) as participants allocated in higher load resistance training groups trained more closely to the requirement for the 1-RM test and presented greater transfer to this outcome, and 2) a combination of both neural and skeletal muscle adaptations derived from higher load resistance training programs resulting in greater effects on muscle strength (38, 63-65). Moreover, the magnitude of change in low vs. high-load resistance training was consistent with that previously reported by Schoenfeld et al. (9) and across all load comparisons favouring the highest loads. Regarding the difference between moderate- and high-load resistance, this

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comparison approached statistical significance (P= .072), and the ES of ~0.3 suggests a possible superior gain with high-load training protocols.

Interestingly, within the limitations of the relative paucity of training studies in which women were subjects, in the low- vs. high-load comparison, men derived greater muscle strength benefits than women, while women improved their strength more than men in comparisons involving moderate-load training. These results may be related to sex differences in prior experience with resistance training and perceived exertion with resistance training sessions as previously suggested (66-68), and possible discrepancies between the prescribed and complied resistance training dosage (35). Moreover, the different time course adaptations between men and women may also account for differential maximal strength gains over prolonged resistance training periods (63, 64). However, we are unable to confirm this due to the short duration of the studies analysed (ranging from 6 to 12 weeks), lack of training variables reported in the studies undertaken by women (32, 33, 35-37, 49-52) and the inability for a more robust narrative concerning studies of longer duration that did not make intensity comparisons in their designs (e.g., see ref. (69)). Regarding the year of publication as a covariate, this variable was used because of the different results on muscle strength between current (5, 9) and past literature (2, 51). These differences are likely to be related to study design, conduct, and control since well-designed studies tend to get smaller effects and reducing bias over the years (70). The larger effect found in the Anderson et al. (2) study also appears to be explained by the low-load resistance training prescription involving more repetitions than any other study (see in Table S2, SDC2, characteristics of included studies), driving the heterogeneity and significance of this

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covariate in the muscle strength model. Furthermore, omitting the studies with a high risk of bias (2, 42, 51) also reduced the significance of this covariate in the model.

The strengths of the present review, network meta-analysis, and meta-regression are: 1) a large number of studies (n= 28) with up to 747 healthy adults; 2); a model involving simultaneous comparison among low-, moderate- and high-load resistance training; and 3) methodological feasibility for comparing different resistance training loads (i.e., statistical transitivity in the network meta-analysis model). In addition, considering previous findings on this topic (9, 19), the present review also extends the results on muscle hypertrophy and strength taking into account different moderators of resistance exercise response such as participants, training status, experimental design, assessed outcomes and training prescription when different resistance training loads are undertaken until volitional failure. However, the present study has some limitations worthy of comment. First, studies included in the present review were mostly low-quality due to concerns regarding the *randomisation process* and *measurement of the outcome*. Nonetheless, we employed a subgroup analysis involving the "best quality" studies to minimise such bias, and the results were maintained with minimal differences in the consistency analysis. Second, the present investigation excluded non-specific muscle strength/performance measures such as maximal voluntary contraction and muscle endurance due to the variability in methods used to assess these outcomes and relevance to our specified outcomes, which would have increased the heterogeneity among studies. Therefore, our results regarding muscle strength should not be extended to non-specific muscle strength tests such as isometric or isokinetic muscle strength tests. Third, although recreationally trained participants were included in our analyses, the strength levels reported in the studies were relatively modest, and certainly, the

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present results cannot be extrapolated for highly strength trained individuals (e.g., bodybuilders, collegiate athletes in structured and supervised strength and conditioning programs and power or strength athletes). Fourth, several studies (4, 5, 33, 37, 38, 43-45, 50, 52) have performed multiple 1-RM testing throughout the study duration (range: 3 to 5 times). This may also be considered an issue masking an even greater difference between low- vs. moderate- and high-load resistance training programs, due to the number of exposures to maximal strength tests and the potential increases in muscle strength not related to the intervention per se (71). Fifth, the assessment of muscle size presents numerous challenges when evaluating training effects with regard to load, movement, range, and type of contraction, all influencing regional hypertrophy and further complicated by location of measurement, technology, and methodology (72). Finally, the present investigation included mostly studies undertaken in young adults (23.4±2.9 years) and should be interpreted with caution when extrapolated to different populations, as exercising until volitional failure is not considered a feasible resistance training prescription for all (73).

In summary, the present study explored the resistance training dosage for muscle hypertrophy and strength in healthy young adults. Our findings suggest that to promote muscle hypertrophy, varying loads of resistance training can be undertaken, and one strategy is to perform the exercise to the point of volitional failure. However, other approaches for maximising recruitment of the motor unit pool maybe as effective, without the issues described above regarding performing high numbers of repetitions with light loads. Thus, a practical application of our results is that a high-load resistance training program (≥80% of 1-RM, or ≤ 8RM) can target both outcomes in shorter periods of training, while a moderate range of repetitions (9RM - 15RM) should be part of the resistance training program for those who do not tolerate exercising

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at higher (i.e., higher intensity or higher resistance) loads, eliciting gains in muscle hypertrophy and muscle strength superior to a low-load program. Furthermore, our results can also be applied to those participants with experience in resistance training, indicating a superior muscle hypertrophy effect undertaking more resistance training sessions.

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ACCEPTED

REFERENCES

1. Delorme TL, Watkins AL. Technics of progressive resistance exercise. *Arch Phys Med Rehabil*. 1948;29(5):263-73.
2. Anderson T, Kearney JT. Effects of three resistance training programs on muscular strength and absolute and relative endurance. *Res Q Exerc Sport*. 1982;53(1):1-7.
3. Campos GE, Luecke TJ, Wendeln HK et al. Muscular adaptations in response to three different resistance-training regimens: specificity of repetition maximum training zones. *Eur J Appl Physiol*. 2002;88(1-2):50-60.
4. Lasevicius T, Ugrinowitsch C, Schoenfeld BJ et al. Effects of different intensities of resistance training with equated volume load on muscle strength and hypertrophy. *Eur J Sport Sci*. 2018;18(6):772-80.
5. Lim C, Kim HJ, Morton RW et al. Resistance Exercise-induced Changes in Muscle Phenotype Are Load Dependent. *Med Sci Sports Exerc*. 2019;51(12):2578-85.
6. Mitchell CJ, Churchward-Venne TA, West DW et al. Resistance exercise load does not determine training-mediated hypertrophic gains in young men. *J Appl Physiol (1985)*. 2012;113(1):71-7.
7. Schoenfeld BJ, Contreras B, Vigotsky AD, Peterson M. Differential Effects of Heavy Versus Moderate Loads on Measures of Strength and Hypertrophy in Resistance-Trained Men. *J Sports Sci Med*. 2016;15(4):715-22.
8. Burd NA, Mitchell CJ, Churchward-Venne TA, Phillips SM. Bigger weights may not beget bigger muscles: evidence from acute muscle protein synthetic responses after resistance exercise. *Appl Physiol Nutr Metab*. 2012;37(3):551-4.

ACCEPTED

9. Schoenfeld BJ, Grgic J, Ogborn D, Krieger JW. Strength and Hypertrophy Adaptations Between Low- vs. High-Load Resistance Training: A Systematic Review and Meta-analysis. *J Strength Cond Res*. 2017;31(12):3508-23.

10. Wernbom M, Augustsson J, Thomee R. The influence of frequency, intensity, volume and mode of strength training on whole muscle cross-sectional area in humans. *Sports Med*. 2007;37(3):225-64.

11. Henneman E. Relation between size of neurons and their susceptibility to discharge. *Science*. 1957;126(3287):1345-7.

12. Adam A, De Luca CJ. Recruitment order of motor units in human vastus lateralis muscle is maintained during fatiguing contractions. *J Neurophysiol*. 2003;90(5):2919-27.

13. Baechle TR, Earle RW. *Essentials of strength training and conditioning*. Human kinetics; 2008.

14. Looney DP, Kraemer WJ, Joseph MF et al. Electromyographical and Perceptual Responses to Different Resistance Intensities in a Squat Protocol: Does Performing Sets to Failure With Light Loads Produce the Same Activity? *J Strength Cond Res*. 2016;30(3):792-9.

15. Miller JD, Lippman JD, Trevino MA, Herda TJ. Neural Drive is Greater for a High-Intensity Contraction Than for Moderate-Intensity Contractions Performed to Fatigue. *J Strength Cond Res*. 2020;34(11):3013-21.

16. Morton RW, Sonne MW, Farias Zuniga A et al. Muscle fibre activation is unaffected by load and repetition duration when resistance exercise is performed to task failure. *J Physiol*. 2019;597(17):4601-13.

ACCEPTED

17. Newton RU, Kraemer WJ. Developing explosive muscular power: Implications for a mixed methods training strategy. *Strength & Conditioning Journal*. 1994;16(5):20-31.

18. Burd NA, West DW, Staples AW et al. Low-load high volume resistance exercise stimulates muscle protein synthesis more than high-load low volume resistance exercise in young men. *PLoS One*. 2010;5(8):e12033.

19. Schoenfeld BJ, Wilson JM, Lowery RP, Krieger JW. Muscular adaptations in low- versus high-load resistance training: A meta-analysis. *Eur J Sport Sci*. 2016;16(1):1-10.

20. Liberati A, Altman DG, Tetzlaff J et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. *BMJ*. 2009;339:b2700.

21. Page MJ, McKenzie JE, Bossuyt PM et al. Mapping of reporting guidance for systematic reviews and meta-analyses generated a comprehensive item bank for future reporting guidelines. *J Clin Epidemiol*. 2020;118:60-8.

22. Furlan AD, Pennick V, Bombardier C, van Tulder M, Editorial Board CBRG. 2009 updated method guidelines for systematic reviews in the Cochrane Back Review Group. *Spine (Phila Pa 1976)*. 2009;34(18):1929-41.

23. American College of Sports M. American College of Sports Medicine position stand. Progression models in resistance training for healthy adults. *Med Sci Sports Exerc*. 2009;41(3):687-708.

24. Sterne JAC, Savovic J, Page MJ et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366:l4898.

25. Higgins J, Thomas J, Chandler J et al. Cochrane handbook for systematic reviews of interventions version 6.0 (updated July 2019). Cochrane. *London, UK*. 2019.

ACCEPTED

26. Shim S, Yoon BH, Shin IS, Bae JM. Network meta-analysis: application and practice using Stata. *Epidemiol Health*. 2017;39:e2017047.

27. Shim SR, Kim SJ, Lee J, Rucker G. Network meta-analysis: application and practice using R software. *Epidemiol Health*. 2019;41:e2019013.

28. Cohen J. Statistical power analysis. *Current directions in psychological science*. 1992;1(3):98-101.

29. Egger M, Davey Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ*. 1997;315(7109):629-34.

30. Rucker G, Schwarzer G, Krahn U, Konig J. Network meta-analysis using frequentist methods. In. R package version 1.2.1.2017.

31. White IR. Multivariate random-effects meta-regression: updates to mvmeta. *The Stata Journal*. 2011;11(2):255-70.

32. Cholewa JM, Rossi FE, MacDonald C et al. The Effects of Moderate- Versus High-Load Resistance Training on Muscle Growth, Body Composition, and Performance in Collegiate Women. *J Strength Cond Res*. 2018;32(6):1511-24.

33. Dinyer TK, Byrd MT, Garver MJ et al. Low-Load vs. High-Load Resistance Training to Failure on One Repetition Maximum Strength and Body Composition in Untrained Women. *J Strength Cond Res*. 2019;33(7):1737-44.

34. Fink J, Kikuchi N, Yoshida S, Terada K, Nakazato K. Impact of high versus low fixed loads and non-linear training loads on muscle hypertrophy, strength and force development. *Springerplus*. 2016;5(1):698.

ACCEPTED

35. Franco CMC, Carneiro M, Alves LTH, Junior GNO, de Sousa JFR, Orsatti FL. Lower-Load is More Effective Than Higher-Load Resistance Training in Increasing Muscle Mass in Young Women. *J Strength Cond Res*. 2019;33 Suppl 1:S152-S8.

36. Herman-Montemayor JR, Hikida RS, Staron RS. Early-Phase Satellite Cell and Myonuclear Domain Adaptations to Slow-Speed vs. Traditional Resistance Training Programs. *J Strength Cond Res*. 2015;29(11):3105-14.

37. Hisaeda H, Miyagawa K, Kuno S, Fukunaga T, Muraoka I. Influence of two different modes of resistance training in female subjects. *Ergonomics*. 1996;39(6):842-52.

38. Jenkins NDM, Miramonti AA, Hill EC et al. Greater Neural Adaptations following High- vs. Low-Load Resistance Training. *Front Physiol*. 2017;8:331.

39. Lasevicius T, Schoenfeld BJ, Silva-Batista C et al. Failure Promotes Greater Muscle Hypertrophy in Low-Load but Not in High-Load Resistance Training. *J Strength Cond Res*. 2019; doi: 10.1519/JSC.0000000000003454. Online ahead of print.

40. Lopes CR, Aoki MS, Crisp AH et al. The Effect of Different Resistance Training Load Schemes on Strength and Body Composition in Trained Men. *J Hum Kinet*. 2017;58:177-86.

41. Mangine GT, Hoffman JR, Gonzalez AM et al. The effect of training volume and intensity on improvements in muscular strength and size in resistance-trained men. *Physiol Rep*. 2015;3(8).

42. Mikkola J, Vesterinen V, Taipale R, Capostagno B, Hakkinen K, Nummela A. Effect of resistance training regimens on treadmill running and neuromuscular performance in recreational endurance runners. *J Sports Sci*. 2011;29(13):1359-71.

ACCEPTED

43. Morton RW, Oikawa SY, Wavell CG et al. Neither load nor systemic hormones determine resistance training-mediated hypertrophy or strength gains in resistance-trained young men. *J Appl Physiol (1985)*. 2016;121(1):129-38.

44. Nobrega SR, Ugrinowitsch C, Pintanel L, Barcelos C, Libardi CA. Effect of Resistance Training to Muscle Failure vs. Volitional Interruption at High- and Low-Intensities on Muscle Mass and Strength. *J Strength Cond Res*. 2018;32(1):162-9.

45. Ozaki H, Kubota A, Natsume T et al. Effects of drop sets with resistance training on increases in muscle CSA, strength, and endurance: a pilot study. *J Sports Sci*. 2018;36(6):691-6.

46. Rana SR, Chleboun GS, Gilders RM et al. Comparison of early phase adaptations for traditional strength and endurance, and low velocity resistance training programs in college-aged women. *J Strength Cond Res*. 2008;22(1):119-27.

47. Schoenfeld BJ, Peterson MD, Ogborn D, Contreras B, Sonmez GT. Effects of Low- vs. High-Load Resistance Training on Muscle Strength and Hypertrophy in Well-Trained Men. *J Strength Cond Res*. 2015;29(10):2954-63.

48. Schoenfeld BJ, Ratamess NA, Peterson MD, Contreras B, Sonmez GT, Alvar BA. Effects of different volume-equated resistance training loading strategies on muscular adaptations in well-trained men. *J Strength Cond Res*. 2014;28(10):2909-18.

49. Schuenke MD, Herman JR, Gliders RM et al. Early-phase muscular adaptations in response to slow-speed versus traditional resistance-training regimens. *Eur J Appl Physiol*. 2012;112(10):3585-95.

ACCEPTED

50. Stefanaki DGA, Dzulkarnain A, Gray SR. Comparing the effects of low and high load resistance exercise to failure on adaptive responses to resistance exercise in young women. *J Sports Sci*. 2019;37(12):1375-80.

51. Stone WJ, Coulter SP. Strength/endurance effects from three resistance training protocols with women. *The Journal of Strength & Conditioning Research*. 1994;8(4):231-4.

52. Van Roie E, Bautmans I, Boonen S, Coudyzer W, Kennis E, Delecluse C. Impact of external resistance and maximal effort on force-velocity characteristics of the knee extensors during strengthening exercise: a randomized controlled experiment. *J Strength Cond Res*. 2013;27(4):1118-27.

53. Vargas S, Petro JL, Romance R et al. Comparison of changes in lean body mass with a strength- versus muscle endurance-based resistance training program. *Eur J Appl Physiol*. 2019;119(4):933-40.

54. Au JS, Oikawa SY, Morton RW, Macdonald MJ, Phillips SM. Arterial Stiffness Is Reduced Regardless of Resistance Training Load in Young Men. *Med Sci Sports Exerc*. 2017;49(2):342-8.

55. Smith RA, Martin GJ, Szivak TK et al. The effects of resistance training prioritization in NCAA Division I Football summer training. *J Strength Cond Res*. 2014;28(1):14-22.

56. Hakkinen K, Pakarinen A. Acute hormonal responses to two different fatiguing heavy-resistance protocols in male athletes. *J Appl Physiol (1985)*. 1993;74(2):882-7.

57. Dankel SJ, Jessee MB, Mattocks KT et al. Training to Fatigue: The Answer for Standardization When Assessing Muscle Hypertrophy? *Sports Med*. 2017;47(6):1021-7.

ACCEPTED

58. Izquierdo M, Ibanez J, Gonzalez-Badillo JJ et al. Differential effects of strength training leading to failure versus not to failure on hormonal responses, strength, and muscle power gains. *J Appl Physiol (1985)*. 2006;100(5):1647-56.

59. Izquierdo-Gabarren M, Gonzalez De Txabarri Exposito R, Garcia-pallares J, Sanchez-medina L, De Villarreal ES, Izquierdo M. Concurrent endurance and strength training not to failure optimizes performance gains. *Med Sci Sports Exerc*. 2010;42(6):1191-9.

60. Testa M, Noakes TD, Desgorces FD. Training state improves the relationship between rating of perceived exertion and relative exercise volume during resistance exercises. *J Strength Cond Res*. 2012;26(11):2990-6.

61. Sale DG. Neural adaptation to resistance training. *Med Sci Sports Exerc*. 1988;20(5 Suppl):S135-45.

62. Coffey VG, Zhong Z, Shield A et al. Early signaling responses to divergent exercise stimuli in skeletal muscle from well-trained humans. *FASEB J*. 2006;20(1):190-2.

63. Hakkinen K. Neuromuscular adaptation during strength training, ageing, detraining, and immobilization. *Crit Rev Phys Rehabil Med*. 1994;14:161-98.

64. Häkkinen K, Komi PV. Electromyographic changes during strength training and detraining. *Med Sci Sports Exerc*. 1983;15(6):455-60.

65. Moritani T. Neural factors versus hypertrophy in the time course of muscle strength gain. *Am J Phys Med Rehabil*. 1979;58(3):115-30.

66. Cotter JA, Garver MJ, Dinyer TK, Fairman CM, Focht BC. Ratings of Perceived Exertion During Acute Resistance Exercise Performed at Imposed and Self-Selected Loads in Recreationally Trained Women. *J Strength Cond Res*. 2017;31(8):2313-8.

ACCEPTED

67. Focht BC. Perceived exertion and training load during self-selected and imposed-intensity resistance exercise in untrained women. *J Strength Cond Res*. 2007;21(1):183-7.

68. Ratamess NA, Faigenbaum AD, Hoffman JR, Kang J. Self-selected resistance training intensity in healthy women: the influence of a personal trainer. *J Strength Cond Res*. 2008;22(1):103-11.

69. Kraemer WJ, Mazzetti SA, Nindl BC et al. Effect of resistance training on women's strength/power and occupational performances. *Med Sci Sports Exerc*. 2001;33(6):1011-25.

70. Pannucci CJ, Wilkins EG. Identifying and avoiding bias in research. *Plast Reconstr Surg*. 2010;126(2):619-25.

71. Spitz RW, Bell ZW, Wong V et al. Strength testing or strength training: considerations for future research. *Physiol Meas*. 2020;41(9):09TR1.

72. Haun CT, Vann CG, Roberts BM, Vigotsky AD, Schoenfeld BJ, Roberts MD. A Critical Evaluation of the Biological Construct Skeletal Muscle Hypertrophy: Size Matters but So Does the Measurement. *Front Physiol*. 2019;10:247.

73. da Silva LXN, Teodoro JL, Menger E et al. Repetitions to failure versus not to failure during concurrent training in healthy elderly men: A randomized clinical trial. *Exp Gerontol*. 2018;108:18-27.

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FIGURE LEGEND

Figure 1. Flow chart of study selection process.

Figure 2. Network geometry of studies examining muscle hypertrophy (n= 25; panel A) and muscle strength (n= 24; panel B). k, number of comparisons; RT, Resistance training.

Figure 3. Standardised mean difference effects between low-, moderate- and high-load resistance training performed until volitional failure on muscle hypertrophy. Overall and subgroup analyses conducted with a network random effects model. Grey and white circles represent study specific estimates based on risk of bias assessment (Low risk, and some concern or high risk of bias, respectively); diamonds represent pooled estimates of random-effects meta-analysis.

Figure 4. Standardised mean difference effects between low-, moderate- and high-load resistance training performed until volitional failure on muscle strength. Overall and subgroup analyses conducted with a network random effects model. Grey and white circles represent study specific estimates based on risk of bias assessment (Low risk, and some concern or high risk of bias, respectively); diamonds represent pooled estimates of random-effects meta-analysis.

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* (SDC1) Table S1. Literature search strategy used for the PubMed database

* (SDC2) Table S2. Study characteristics: author and year, sample, experimental design, resistance training prescription and outcomes assessed.

* (SDC3) Figure S1. Individual risk of bias assessment at outcome level for A) muscle hypertrophy and B) muscle strength. Green circles, low risk; yellow circles, some concerns; red circles, high risk of bias.

* (SDC4) Table S3. Network univariate meta-regression models for muscle hypertrophy and muscle strength.

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Figure 1

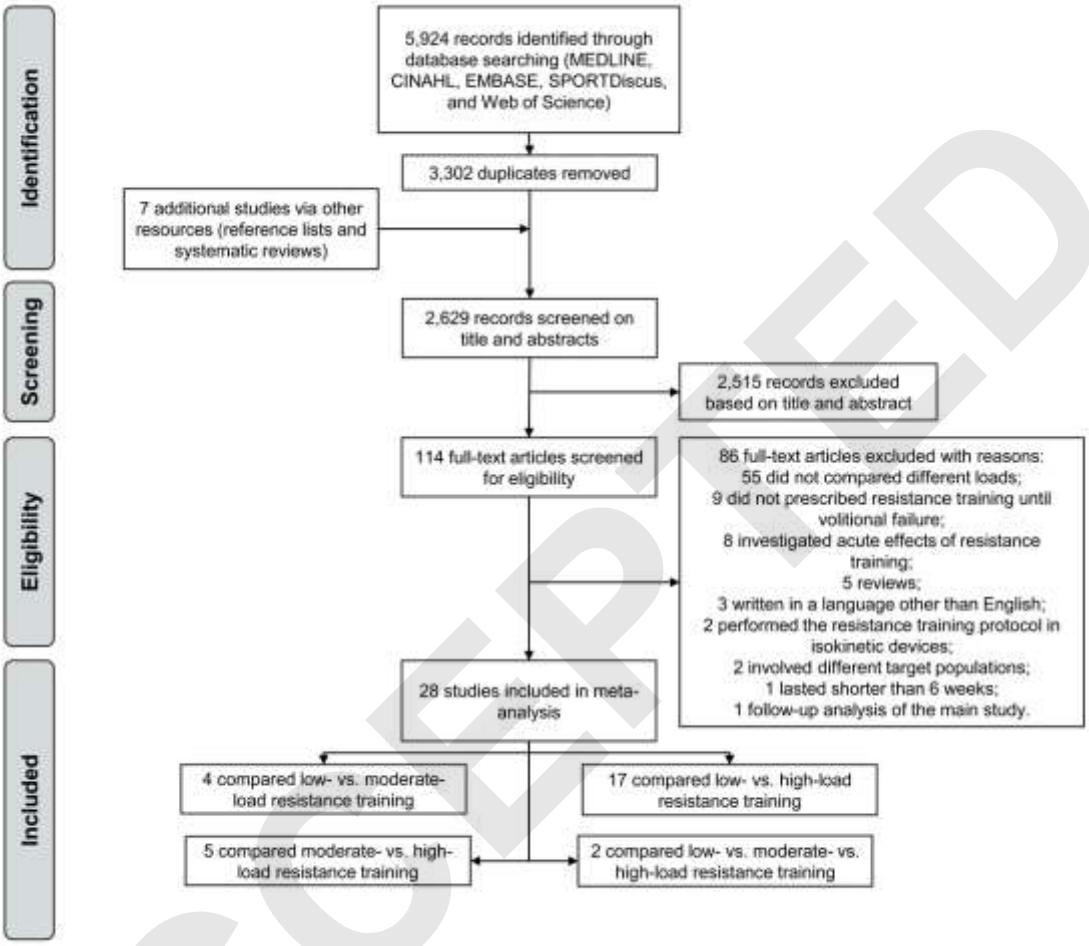

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Figure 2

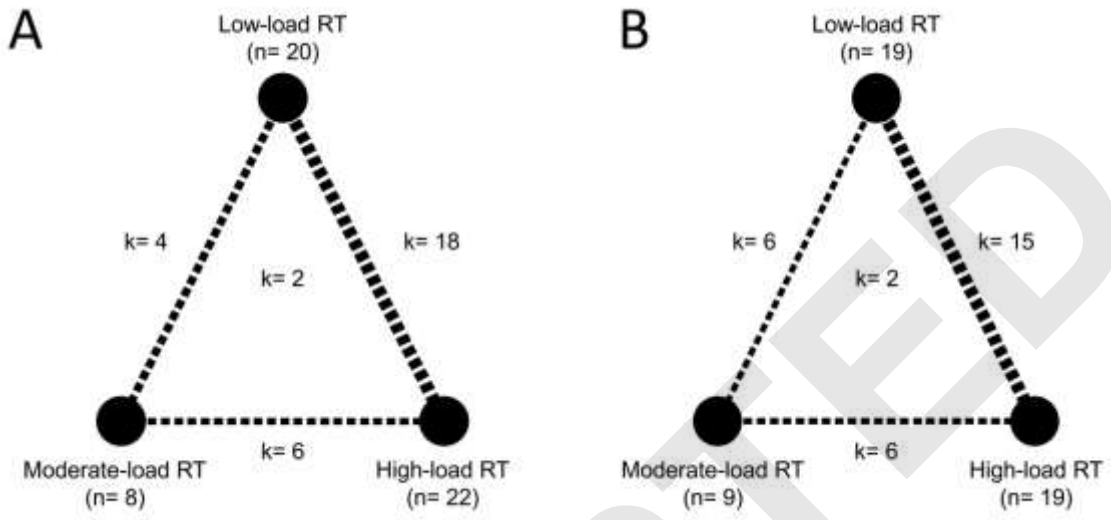

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Figure 3

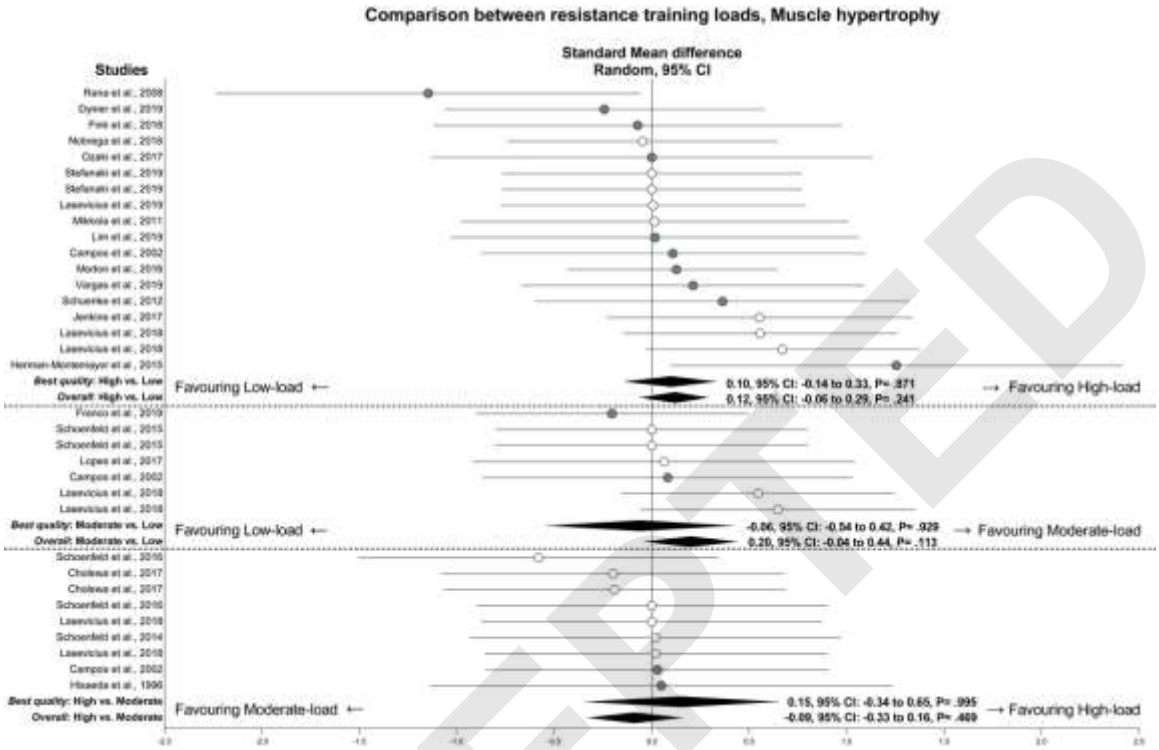

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Figure 4

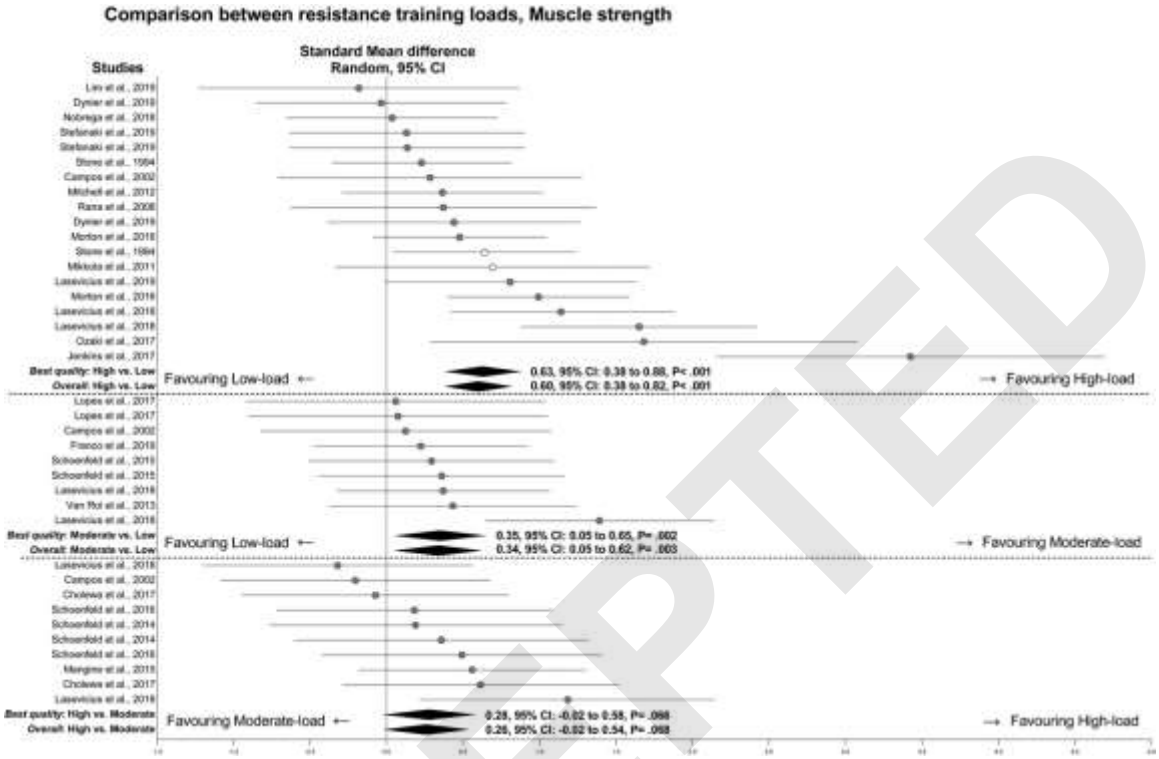

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Table 1. Risk of bias of included studies.

Outcome	Randomisation process	Deviation from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall bias
Muscle hypertrophy, n= 24						
Low risk	0	24 (100%)	24 (100%)	13 (54.2%)	24 (100%)	0
Some concerns	23 (95.8%)	0	0	8 (33.3%)	0	21 (87.5%)
High risk	1 (4.2%)	0	0	3 (12.5%)	0	3 (12.5%)
Muscle strength, n= 23						
Low risk	0	23 (100%)	23 (100%)	0	23 (100%)	0
Some concerns	20 (87.0%)	0	0	23 (100%)	0	20 (87.0%)
High risk	3 (13.0%)	0	0	0	0	3 (13.0%)

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Table 2. Network meta-analysis consistency models for muscle hypertrophy and muscle strength in studies comparing low-, moderate-, and high-load resistance training in healthy adults.

Outcome	Comparisons		k	Sample	Pooled SMD	95% CI	P-value	Best intervention probability
Muscle hypertrophy	High vs. Low	All	19	347	0.12	-0.06 to 0.29	.241	*Overall analysis:*
		Best-quality	12	274	0.10	-0.14 to 0.33	.871	84.5% for moderate-load resistance
	Moderate vs. Low	All	7	128	0.20	-0.04 to 0.44	.113	training;
		Best-quality	2	88	-0.06	-0.54 to 0.42	.929	*Best quality:*
	High vs. Moderate	All	9	107	-0.09	-0.33 to 0.16	.469	75.8% for high-load resistance
		Best-quality	2	51	0.15	-0.34 to 0.65	.995	training
Muscle strength	High vs. Low	All*	19	403	0.60	0.38 to 0.82	<.001	*Overall analysis:*
		Best-quality	16	325	0.63	0.38 to 0.88	<.001	98.2% for high-load resistance
	Moderate vs. Low	All*	9	152	0.34	0.05 to 0.62	.003	training;
		Best-quality	9	152	0.35	0.05 to 0.65	.002	*Best quality:*
	High vs. Moderate	All*	10	125	0.26	-0.02 to 0.54	.068	98.2% for high-load resistance
		Best-quality	10	125	0.28	-0.02 to 0.58	.066	training

*, Adjustment after sensitivity analysis omitting one study at a time; k, Number of comparisons.

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Table 3. Network meta-regression models for muscle hypertrophy and muscle strength.

Comparison	Covariates	Range	Coef±SE	95% CI	P-value
Muscle hypertrophy					
High vs. Low	Year of publication	2002 to 2019	0.05 ± 0.05	-0.04 to 0.14	.244
	Experimental design	Between- vs. Within-group	0.00 ± 0.33	-0.64 to 0.64	.999
	Sex	Women vs. Men	0.22 ± 0.40	-0.64 to 1.01	.583
	Training status	Untrained vs. Recreationally trained	-1.38 ± 0.65	-2.67 to -0.11	**.033**
	Number of sessions	12 to 48	0.01 ± 0.01	-0.02 to 0.05	.518
	Assessed limb	Lower- vs. Upper-body	0.03 ± 0.25	-0.46 to 0.52	.902
	Prescription method	%1-RM vs. RM's	1.02 ± 0.70	-0.35 to 2.40	.145
Moderate vs. Low	Year of publication	2002 to 2019	0.02 ± 0.05	-0.07 to 0.11	.627
	Experimental design	Between vs. Within-group	-0.66 ± 0.86	-2.34 to 1.03	.447
	Sex	Women vs. Men	0.70 ± 0.76	-0.80 to 2.19	.360
	Training status	Untrained vs. Recreationally trained	-1.15 ± 0.82	-2.76 to 0.46	.161
	Number of sessions	16 to 24	0.09 ± 0.04	0.01 to 0.16	**.031**
	Assessed limb	Lower- vs. Upper-body	-0.11 ± 0.31	-0.70 to 0.49	.725
	Prescription method†	%1-RM vs. RM's	-	-	-
High vs. Moderate	Year of publication	1996 to 2018	-0.03 ± 0.03	-0.09 to 0.03	.342
	Experimental design	Between vs. Within-group	0.36 ± 0.70	-1.00 to 1.74	.604
	Sex	Women vs. Men	0.48 ± 0.66	-0.82 to 1.78	.471
	Training status	Untrained vs. Recreationally trained	0.23 ± 0.65	-1.05 to 1.51	.721
	Number of sessions	16 to 33	0.08 ± 0.04	0.01 to 0.15	**.045**
	Assessed limb	Lower- vs. Upper-body	-0.14 ± 0.31	-0.74 to 0.46	.651
	Prescription method†	%1-RM vs. RM's	-	-	-
Muscle strength					
High vs. Low	Year of publication	1982 to 2019	-0.11 ± 0.05	-0.20 to -0.15	**.023**
	Experimental design	Between vs. Within-group	-0.47 ± 0.58	-1.60 to 0.66	.411
	Sex	Women vs. Men	1.03 ± 0.50	0.06 to 2.00	**.037**
	Training status	Untrained vs. Recreationally trained	0.73 ± 1.15	-1.53 to 3.00	.526
	Number of sessions	12 to 48	-0.01 ± 0.03	-0.06 to 0.04	.715
	Assessed limb	Lower- vs. Upper-body	0.28 ± 0.45	-0.60 to 1.16	.535
	Prescription method	%1-RM vs. RM's	-1.79 ± 1.15	-4.05 to 0.47	.121
Moderate vs. Low	Year of publication	2002 to 2019	-0.06 ± 0.23	-0.52 to 0.39	.787
	Experimental design†	Between vs. Within-group	-	-	-
	Sex	Women vs. Men	-0.95 ± 4.26	-9.30 to 7.40	.823
	Training status	Untrained vs. Recreationally trained	1.65 ± 3.68	-5.56 to 8.86	.654

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	Number of sessions	16 to 27	-0.08 ± 0.07	-0.23 to 0.06	.263
	Assessed limb	Lower- vs. Upper-body	-0.20 ± 0.49	-1.16 to 0.76	.677
	Prescription method	%1-RM vs. RM's	-2.34 ± 4.26	-10.7 to 6.01	.583
High vs. Moderate	Year of publication	2002 to 2017	0.04 ± 0.23	-0.40 to 0.48	.848
	Experimental design†	Between vs. Within-group	-	-	-
	Sex	Women vs. Men	-1.99 ± 4.06	-9.94 to 5.98	.625
	Training status	Untrained vs. Recreationally trained	0.91 ± 3.37	-5.70 to 7.53	.786
	Number of sessions	16 to 38	-0.07 ± 0.07	-0.20 to 0.06	.283
	Assessed limb	Lower- vs. Upper-body	-0.48 ± 0.53	-1.52 to 0.55	.360
	Prescription method	%1-RM vs. RM's	-1.02 ± 4.04	-8.95 to 6.90	.800

†, Collinearity detected given the insufficient number of observations. %1-RM, Percentage of one repetition maximum; RM's, Repetitions maximum.

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(SDC1) Table S1. Literature search strategy used for the PubMed database

#1" Search "strength training to failure" [title/abstract] OR "strength training to muscle failure" [title/abstract] OR "muscle failure" [title/abstract] OR "muscular failure" [title/abstract] OR "set to failure" [title/abstract] OR "sets to failure" [title/abstract] OR "until failure" [title/abstract] OR "until volitional failure" [title/abstract] OR "protocols to failure" [title/abstract] OR "strength training to exhaustion" [title/abstract] OR "strength training to muscle exhaustion" [title/abstract] OR "set to exhaustion" [title/abstract] OR "sets to exhaustion" [title/abstract] OR "until exhaustion" [title/abstract] OR "until volitional exhaustion" [title/abstract] OR "protocols to exhaustion" [title/abstract] OR "strength training to fatigue" [title/abstract] OR "strength training to muscle fatigue" [title/abstract] OR "muscular fatigue" [title/abstract] OR "muscle fatigue" [title/abstract] OR "set to fatigue" [title/abstract] OR "sets to fatigue" [title/abstract] OR "until fatigue" [title/abstract] OR "until volitional fatigue" [title/abstract] OR "protocols to fatigue" [title/abstract] OR "repetitions maximum" [title/abstract] OR "repetition maximum" [title/abstract] OR "resistance training to failure" [title/abstract] OR "resistance training to muscle failure" [title/abstract] OR "resistance training to exhaustion" [title/abstract] OR "resistance training to muscle exhaustion" [title/abstract] OR "resistance training to fatigue" [title/abstract] OR "resistance training to muscle fatigue" [title/abstract] OR "high-load" [title/abstract] OR "high load" [title/abstract] OR "low-load" [title/abstract] OR "low load" [title/abstract]
#2" Search "muscle mass" [title/abstract] OR "muscular size" [title/abstract] OR hypertrophy [title/abstract] OR "muscle hypertrophy" [title/abstract] OR "cross-sectional area" [title/abstract] OR "muscle thickness" [title/abstract] OR "one maximum repetition" [title/abstract] OR "1-RM" [title/abstract] OR "maximal strength" [title/abstract].
#1 AND #2

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(SDC2) Table S2. Study characteristics: author and year, sample, experimental design, resistance training prescription and outcomes assessed.

Author, year	Sample	Experimental Design	Resistance training prescription	Resistance training volume	Outcomes
Anderson et al., 1982 (2)	n= 43 age: 20.7±1.8 yrs Untrained men	Between-subject	27 sessions (9wk, 3x.wk^{-1}); 1 exercise; High load: 3 sets of 6-8RM (n= 15) Low load: 3 sets of 30-40RM (n= 16) Low load: 3 sets of 100-150RM (n= 12)	NR	Bench press 1-RM
Stone & Coulter, 1994 (51)	n= 50 age: 23.1±3.5 yrs Untrained women	Between-subject	27 sessions (9wk, 3x.wk^{-1}); 5 exercises; High load: 3 sets of 6-8RM (n= 17) Low load: 2 sets of 15-20RM (n= 16) Low load: 1 set of 30-40RM (n= 17)	NR	Bench press 1-RM; Squat 1-RM
Hisaeda et al., 1997 (37)	n= 11 age: 19.8±1.8 and 20.4±1.3 yrs Untrained women	Between-subject	24 sessions (8wk, 3x.wk^{-1}); 1 exercise; High load: 3-5 sets of 4.5RM (n= 6) Moderate load: 3-5 sets of 8-20RM (n= 5)	Knee extension volume[c] High load: range from 2.2±0.5 to 3.6±0.6 kJ Moderate load: range from 3.5±0.4 to 4.7±0.7 kJ	Quadriceps femoris CSA (MRI)
Campos et al., 2002 (3)	n= 32 age: 22.5±5.8 yrs Untrained men	Between-subject	16 sessions (8wk, 2x.wk^{-1}); 3 exercises; High load: 4 sets of 3-5RM (n= 9) Moderate load: 3 sets 9-11RM (n= 11) Low load: 3 sets of 20-28RM (n= 7)	NR	Squat 1-RM; Leg press 1-RM; Knee Extension 1-RM; Vastus lateralis Biopsy

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Rana et al., 2008 (46)	n= 34 age: 21.1±2.7 yrs Untrained men	Between-subject	17 sessions (6wk, 2-3x.wk^{-1}); 3 exercises; High load: 3 sets of 6-10RM (n= 9) Low load: 3 sets of 20-30RM (n= 8)	NR	Squat 1-RM; Leg press 1-RM; Knee extension 1-RM; Fat-free mass (BodPod)
Mikkola et al., 2011 (42)	n= 27 age: 34±9 and 36±6 yrs Active untrained men	Between-subject	? sessions (14wk, 1-2x.wk^{-1}); 8 exercises; High load: 3 sets of 4-6RM (n= 11) Low load: 3 sets of 40-50RM (n= 6)	Leg press volume[a] High load: range from 1,297±137 to 1,615±213 kg; Low load: NR	Leg press 1-RM; Fat-free mass (Skin fold)
Mitchell et al., 2012 (6)	n= 18 age: 21±1 yrs Active untrained men	Within-subject	30 sessions (10wk, 3x.wk^{-1}); 1 exercise; High load: 3 sets at 80% of 1-RM (n= 18) Low load: 3 sets of 30% of 1-RM (n= 18)	NR	Knee extension 1-RM; Quadriceps femoris volume (MRI); Vastus lateralis Biopsy
Schuenke et al., 2012 (49)	n= 34 age: 21.1±2.7 yrs Untrained women	Between-subject	17 sessions (6wk, 2-3x.wk^{-1}); 3 exercises; High load: 3 sets of 6-10RM Low load: 3 sets of 20-30RM	NR	Vastus lateralis Biopsy
Van Roie et al., 2013 (52)	n= 36 age: 21.4±1.8 and 22.2±2.7 yrs Untrained men and women	Between-subject	27 sessions (9wk, 3x.wk^{-1}); 1 exercise; Moderate load: 1 set of 10-12RM (n= 12) Low-load: 1 set of 60 reps at 20-25% of 1-RM plus 10-12 reps at 40% of 1-RM (n= 12)	NR	Knee extension 1-RM

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Schoenfeld et al., 2014 (48)	n= 20 age: 23.2±2.7 yrs ~~Men with~~ experience in resistance training	Between-subject	24 sessions (8wk, 3x.wk^{-1}); 9 exercises; High load: 7 sets of 3RM (n= 10) Moderate load: 3 sets of 10RM (n= 10)	Bench press volume[a] High load: 4,140±49 kg; Moderate load: 3,693±47 Squat volume[a] High volume: 5,751±68 kg; Moderate volume: 6,625±85 kg	Bench press 1-RM; Squat 1-RM; Elbow flexor MT (Ultrasound)
Herman-Montemayor et al., 2015 (36)	n= 34 age: 21.1±2.7 yrs Untrained women	Between-subject	21 sessions (8wk, 2-3x wk^{-1}); 3 exercises; High load: 3 sets of 6-10RM (n= 9) Low load: 3 sets of 20-30RM (n= 6)	NR	Vastus lateralis Biopsy
Mangine et al., 2015 (41)	n= 29 age: 24.0±2.7 and 24.7±3.4 yrs Men with experience in resistance training	Between-subject	38 sessions (10wk, 2-4x.wk^{-1}); 17 exercises; High load: 4 sets of 3-5RM (n= 15) Moderate load: 4 sets of 10-12RM (n= 14)	Bench press volume[a] High load: 2,757±696 kg; Moderate load: 4,412±729 kg	Bench press 1-RM
Schoenfeld et al., 2015 (47)	n= 24 age: 23.3±? yrs Men with experience in resistance training	Between-subject	24 sessions (8wk, 3x.wk^{-1}); 7 exercises; Moderate load: 3 sets of 8-12RM (n= 12) Low load: 3 sets of 25-35RM (n= 12)	NR	Bench press 1-RM; Squat 1-RM; Elbow flexor MT (Ultrasound); Elbow extensor MT (Ultrasound)
Fink et al., 2016 (34)	n= 21 age: 23.1±2.4 and 23.4±3.0 yrs Untrained men	Between-subject	24 sessions (8wk, 3x.wk^{-1}); 1 exercise; High load: 3 sets at 80% of 1-RM (n= 7)	Biceps curl volume[b] High load: 15.3±1.6 repetitions; Low load: 75.3±12.6 repetitions	Arm CSA (MRI)

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	athletes		Low load: 3 sets at 30% of 1-RM (n= 7)		
Morton et al., 2016 (43)	n= 49 age: 23±1 yrs Men with experience in resistance training	Between-subject	48 sessions (12wk, 4x.wk^{-1}); 10 exercises; High load: 3 sets of 8-12RM* (n= 27) Low load: 3 sets of 20-25RM (n= 29)	Total volume per session[a] High load: 14,805±592 kg; Low load: 23,969±901 kg	Bench press 1-RM; Leg press 1-RM; Knee extension 1-RM; Vastus lateralis Biopsy
Schoenfeld et al., 2016 (7)	n= 19 age: 23.2±4.2 yrs Men with experience in resistance training	Between-subject	24 sessions (8wk, 3x.wk^{-1}); 7 exercises; High load: 7 sets of 3RM (n= 10) Moderate load: 3 sets of 8-12RM (n= 9)	Total volume[a] High load: 25,867±3,731 kg; Moderate load: 56,049±11,101 kg	Bench press 1-RM; Squat 1-RM; Elbow flexor MT (Ultrasound); Elbow extensor MT (Ultrasound); Vastus lateralis MT (Ultrasound)
Cholewa et al., 2017 (32)	n= 20 age: 20.3±1.5 yrs Untrained women	Between-subject	33 sessions (11wk, 3x.wk^{-1}); 13 exercises; High load: 4 sets at 5-7 RM (n= 10) Moderate load: 2 sets at 10-14RM (n= 10)	Squat total volume[a] High load: range from 1,234 to 1,600 kg; Moderate load: range from 947 to 1,262kg.	Squat 1-RM; Arm and thigh CSA (Skinfold circumference)
Jenkins et al., 2017 (38)	n= 26 age: 23.1±4.7 yrs Untrained men	Between-subject	18 sessions (6wk, 3x.wk^{-1}); 1 exercise; High load: 3 sets at 80% of 1-RM (n= 13) Low load: 3 sets at 30% of 1-RM (n= 13)	Knee extension volume[a] High load: 38,825±3,003 kg; Low load: 41,170±4,327 kg	Knee extension 1-RM; Quadriceps femoris MT (Ultrasound)
Lopes et al., 2017 (40)	n= 16 age: 24.6±5.8 and 28.6±6 yrs	Between-subject	24 sessions (6wk, 4x.wk^{-1}); 15 exercises;	Total volume[a] Moderate load: range from 89,029±15,297 to 106,802±14,903 kg;	Bench press 1-RM; Squat 1-RM;

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	Men with experience in resistance training		Moderate load: 6 sets of 10RM (n= 8) Low load: 3 sets of 20RM (n= 8)	Low load: range from 80,460±10,721 to 97,131±11,109 kg	Fat-free mass (Skin fold)
Ozaki et al., 2017 (45)	n= 9 age: 26±1 yrs Untrained men	Within-subject	22 sessions (8wk, 2-3x.wk^{-1}); 1 exercise; High load: 3 sets of 80% of 1-RM (n= 6) Low load: 3 sets of 30% of 1-RM (n= 6)	Elbow flexion volume per set[b] High load: range from 8.8±1.3 to 5.1±1.2 repetitions; Low load: range from 104.3±22.3 to 31.8±5.6 repetitions	Elbow flexors 1-RM; Arm CSA (MRI)
Lasevicius et al., 2018 (4)	n= 30 age: 24.5±2.4 yrs Active untrained men	Within-subject	24 sessions (12wk, 2x.wk^{-1}); 2 exercises; High load: 4 sets at 80% of 1-RM (n= 10) Moderate load: 5 sets at 60% of 1-RM (n= 10) Low load: 4 sets at 40% of 1-RM (n= 10) Low load: 3 sets at 20% of 1-RM (n= 30)	Leg press volume[a] High load: 160,005±36,054 kg; Moderate load: 160,900±30,997 kg; Low load: 153,946±28,285 and 163,043±34,695 kg Elbow flexion volume[a] High load: 20,543±3,480 Moderate load: 20,822±3,644 Low load: 21,919±3,028 and 21,674±2953 kg	Leg press 1-RM; Elbow flexors 1-RM; Vastus lateralis CSA (Ultrasound); Elbow flexor CSA (Ultrasound)
Nobrega et al., 2018 (44)	n= 32 age: 23.0±3.6 yrs Untrained men	Within-subject	24 sessions (12wk, 2x.wk^{-1}); 1 exercise; High load: 3 sets at 80% of 1-RM (n= 14) Low load: 3 sets at 30% of 1-RM (n= 13)	Knee extension volume[a] High load: range from 12,795±3,654 to 13,899±3,236 kg; Low load: range from 9,923±3,144 to 11,191±3,346 kg	Knee extension 1-RM; Vastus lateralis CSA (Ultrasound)
Dinyer et al., 2019 (33)	n= 23 age: 21.2±2.2 yrs Untrained women	Between-subject	24 sessions (12wk, 2x.wk^{-1}); 4 exercises; High load: 2-3 sets at 80% of 1-RM (n=	Knee extension volume[a] High load: 14,845 kg; Low load: 11,755 kg	Knee extension 1-RM; Seated military press 1-RM; Bone- and fat-free mass (DXA)

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			12) Low load: 2-3 sets at 30% of 1-RM (n= 11)	Seated military press[a] High load: 4,366 kg; Low load: 4,030 kg	
Franco et al., 2019 (35)	n= 32 age: 23.0±3.3 and 24.3±4.8 yrs Untrained women	Between-subject	18 sessions (9wk, 2x.wk^{-1}); 3 exercises; Moderate Load: 3 sets of 8-10RM (n= 18) Low load: 3 sets of 30-35RM (n=14)	Total volume[a] High load: 2,993±561 kg; Low load: 5,190±731 kg	Knee extension 1-RM; Leg bone- and fat-free mass (DXA)
Lasevicius et al., 2019 (39)	n= 32 age: 23.8±4.9 and 24.3±4.8 yrs Active untrained men	Within-subject	16 sessions (8wk, 2x.wk^{-1}); 1 exercise; High load: 3 sets at 80% of 1-RM (n= 13) Low load: 3 sets at 30% of 1-RM (n= 12)	Knee extension volume[a] High load: 34,853±8,020 kg Low load: 34,576±16,372 kg	Knee extension 1-RM; Quadriceps femoris CSA (Ultrasound)
Lim et al., 2019 (5)	n= 21 age: 23±1 and 24±2 yrs Untrained men	Between-subject	30 sessions (10wk, 3x.wk^{-1}); 3 exercises; High load: 3 sets at 80% of 1-RM (n= 7) Low load: 3 sets at 30% of 1-RM (n= 7)	Knee extension volume[a] High load: 1,117±463 kg Low load: 1,571±476 kg	Knee extension 1-RM; Vastus lateralis Biopsy
Stefanaki et al., 2019 (50)	n= 13 age: 29.7±4.7 yrs Untrained women	Within-subject	12 sessions (6wk, 2x.wk^{-1}); 2 exercises; High load: 1 set at 80% of 1-RM (n= 13) Low load: 1 set at 30% of 1-RM (n=13)	Elbow flexors volume[a] High load: range from 274 to 459 kg; Low load: range from 459 to 874 kg Knee extension[a] High load: 97 – 123 kg; Low load: 199 – 255 kg	Elbow flexors 1-RM; Knee extension 1-RM; Elbow flexor MT (Ultrasound); Vastus lateralis MT (Ultrasound)

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Vargas et al., 2019 (53)	n= 25 age:28.4±6.4 yrs Men with experience in resistance training	Between-subject	32 sessions (8wk, 4x.wk^{-1}); 16 exercises; High load: 3 sets of 6-8RM (n= 10) Low load: 3 sets of 20-25RM (n= 10)	NR	Lean body mass (DXA)

*, Corresponding 75-90% of 1-RM as reported by authors; [a], Product of sets, repetitions and load; [b], Average total number of repetitions; [c], Product of training load, distance and repetitions; 1-RM, one repetition maximum; CSA, cross-sectional area; DXA, dual-energy X-ray absorptiometry; MRI, magnetic resonance imaging; MT, muscle thickness; wk, weeks.

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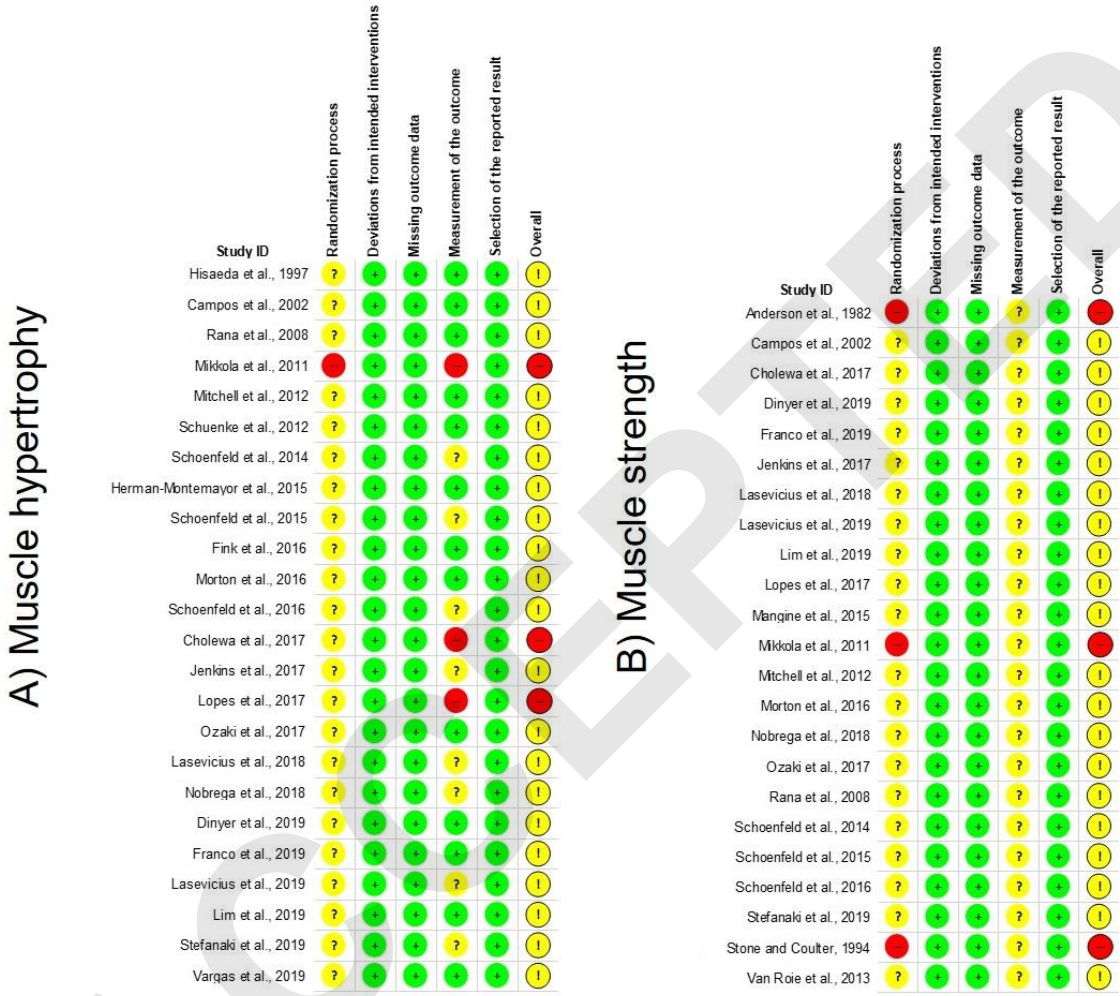
(SDC3) Figure S1. Individual risk of bias assessment at outcome level for A) muscle hypertrophy and B) muscle strength. Green circles, low risk; yellow circles, some concerns; red circles, high risk of bias.



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(SDC4) Table S3. Network univariate meta-regression models for muscle hypertrophy and muscle strength.

Comparison	Covariates	Range	Overall	
			Coef±SE	P-value
Muscle hypertrophy				
High vs. Low	Year of publication	2002 to 2019	0.02 ± 0.02	.438
	Experimental design	Between- vs. Within-group	0.12 ± 0.19	.519
	Sex	Women vs. Men	0.11 ± 0.22	.610
	Training status	Untrained vs. Recreationally trained	-0.08 ± 0.23	.726
	Number of sessions	12 to 48	0.00 ± 0.01	.710
	Assessed limb	Lower- vs. Upper-body	0.00 ± 0.23	.986
	Prescription method	%1-RM vs. RM's	-0.13 ± 0.19	.490
Moderate vs. Low	Year of publication	2002 to 2019	0.02 ± 0.03	.454
	Experimental design	Between vs. Within-group	0.28 ± 0.26	.278
	Sex	Women vs. Men	0.15 ± 0.26	.567
	Training status	Untrained vs. Recreationally trained	-0.11 ± 0.26	.677
	Number of sessions	16 to 24	0.02 ± 0.02	.370
	Assessed limb	Lower- vs. Upper-body	0.07 ± 0.27	.791
	Prescription method	%1-RM vs. RM's	-0.29 ± 0.26	.265
High vs. Moderate	Year of publication	1996 to 2018	0.00 ± 0.02	.970
	Experimental design†	Between vs. Within-group	0.15 ± 0.28	.585
	Sex	Women vs. Men	0.04 ± 0.27	.884
	Training status	Untrained vs. Recreationally trained	-0.02 ± 0.27	.927
	Number of sessions	16 to 33	0.02 ± 0.02	.462
	Assessed limb	Lower- vs. Upper-body	0.07 ± 0.26	.798
	Prescription method	%1-RM vs. RM's	-0.15 ± 0.28	.579
Muscle strength				
High vs. Low	Year of publication	1982 to 2019	-0.05 ± 0.02	**.021**
	Experimental design	Between vs. Within-group	-0.13 ± 0.42	.754
	Sex	Women vs. Men	0.68 ± 0.41	.097
	Training status	Untrained vs. Recreationally trained	-0.20 ± 0.42	.644
	Number of sessions	12 to 48	0.00 ± 0.02	.827
	Assessed limb	Lower- vs. Upper-body	0.30 ± 0.39	.446
	Prescription method	%1-RM vs. RM's	0.08 ± 0.38	.826
Moderate vs. Low	Year of publication	2002 to 2019	-0.02 ± 0.04	.538
	Experimental design	Between vs. Within-group	0.42 ± 0.41	.306
	Sex	Women vs. Men	0.49 ± 0.48	.315
	Training status	Untrained vs. Recreationally trained	-0.27 ± 0.39	.492
	Number of sessions	16 to 27	0.01 ± 0.03	.906
	Assessed limb	Lower- vs. Upper-body	0.01 ± 0.43	.984
	Prescription method†	%1-RM vs. RM's	-0.44 ± 0.39	.267
High vs. Moderate	Year of publication	2002 to 2017	0.02 ± 0.04	.525
	Experimental design	Between vs. Within-group	0.55 ± 0.52	.287
	Sex	Women vs. Men	-0.19 ± 0.36	.593
	Training status	Untrained vs. Recreationally trained	-0.07 ± 0.31	.811

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Number of sessions	16 to 38	-0.00 ± 0.03	.965
Assessed limb	Lower- vs. Upper-body	-0.29 ± 0.33	.384
Prescription method†	%1-RM vs. RM's	-0.52 ± 0.49	.288

%1-RM, Percentage of one repetition maximum; RM's, Repetitions maximum.

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